	TOTAL FOR QUESTION 5	**7**
6(a)	Faraday's law of electromagnetism states that the (induced) e.m.f. is directly proportional to the rate of change of magnetic flux linkage.	**B1**
6(b)(i)	$\varepsilon = \frac{d\phi}{dt} = \frac{\Delta\phi}{\Delta t}$ $\Delta\phi = \varepsilon \times \Delta t$ $= 0.55 \times 14 = 7.7$ Wb Accepted units include: Wb / Vs / Tm^2	**A1** **A1**
6(b)(ii)	$\phi = BA$ $A = \frac{\phi}{B} = \frac{7.7}{38 \times 10^{-6}}$ $= 2.0 \times 10^5$ m^2 **Note:** Allow ECF from **(b)(i)**	**A1**
6(b)(iii)	**Method 1:** Area = speed x time x width $v = \frac{A}{t \times \text{width}}$ $= \frac{2.0 \times 10^5}{14 \times 68}$ $= 210$ m s^{-1}	**C1** **A1**

Qns	Answer	Marks
	OR **Method 2:** $\varepsilon = \frac{d\phi}{dt}$ $\varepsilon = BLv$ **C1** $0.55 = (38 \times 10^{-6})(68)v$ $v = 213 \text{ m s}^{-1}$ $= 210 \text{ m s}^{-1}$ (2 s.f.) **Note:** Allow ECF from **(b)(ii)**	**OR** **C1** **A1**
6(b)(iv)	As the aircraft travels through the magnetic field with a velocity perpendicular to the field lines, charges in the wing experience a magnetic force. Using Fleming's left-hand rule, magnetic forces act on positive (negative) charges, causing them to move toward wingtip P (Q). As positive (negative) charges accumulate at P (Q) wingtip P will be at a higher potential than Q. sp	**B1** **A1**
6(c)(i)	<table><tr><th>Resistor</th><th>r.m.s. voltage, $V_{r.m.s.}$ / V</th><th>peak voltage, V_0 / V</th></tr><tr><td>R_1</td><td>$\frac{16}{\sqrt{2}}$ = 11 (or 11.3)</td><td>16</td></tr><tr><td>R_2</td><td>$\frac{16}{2}$ = 8.0</td><td>16</td></tr></table>	**A1**
6(c)(ii)	The peak voltages are the same because when the diode is forward-biased, its resistance is close to zero, giving both resistors the same peak potential difference. The r.m.s. voltages for both resistors are different because the branch with resistor R_2 and the diode undergoes half-wave rectification, unlike the branch with only resistor R_1.	**B1** **B1**
	TOTAL FOR QUESTION 6	**11**
7(a)	Half-life is the average time taken for half the original number of radioactive nuclei to decay.	**B1**